

PROFILE

Cardiff engineering student with international internship experience at Centre of Robotics. Practical work spans CAD, physical prototyping, sensor integration and experimental testing, alongside a contribution to a published scoping review.

EDUCATION

Cardiff University 2026 – present
BEng Mechanical and Electrical Engineering with a Year in Industry

University of Bristol 2025 – 2026
Foundation Year in Science, Engineering and Mathematics — CertHE, completed 2026

ENGINEERING EXPERIENCE

Robotics Engineer Intern - Centre of Robotics Jun – Sep 2026
São Paulo

- Developed a manual two-jaw gripper in Fusion 360 and produced two 3D-printed prototypes to evaluate component geometry, assembly and jaw movement.
- Participated in hands-on robot interaction and demonstrations involving platforms including Spot.
- Gained introductory exposure to ROS 2 and NVIDIA Isaac, connecting commands, state feedback and simulation with physical hardware. Research collaboration remains ongoing.

Mechanical & Electrical Team Member — Bristol Racing Sep 2025 – Jul 2026
➤ Contributed to a scaled RC vehicle within the Formula Student team : drivetrain packaging, steering concepts, CAD layouts and wiring, considering laser-cut and 3D-printed part assembly.

Junior Analyst — Bristol Academic Research Society Sep 2025 – Jul 2026
➤ Contributed to the published *Multifunctional Self-Healing Structural Composites for Load Bearing Satellite Applications : A Qualitative Scoping Review*; reviewed literature and assessed material-performance trade-offs for satellite structures.

SELECTED PROJECTS

Temperature & Thermal Imaging Monitoring System — Built and tested an Arduino system combining DS18B20 point sensing, an AMG8833 thermal array, LCD and LED alerts. Investigated 30 °C threshold response, stability and distance sensitivity; produced a dissertation and poster.

Smart Lamp — Assembled and tested an Arduino-based lamp with motion and sound sensing; investigated sensor-triggered LED responses, wiring and power delivery.

PROFESSIONAL EXPERIENCE

Programmatic Advertising Roles — Global Media Organisations May 2021 – Oct 2025
Led campaign delivery across global media platforms, analysed performance and resolved operational issues; coordinated technical and commercial stakeholders to meet delivery deadlines.

TECHNICAL SKILLS

CAD : Fusion 360 — CAD modelling & assemblies; Autodesk Inventor — CAD modelling.

Systems : ROS 2 & NVIDIA Isaac — research-lab exposure; Arduino — embedded project development.

Practical & analytical : 3D printing, sensor integration, experimental testing, Excel data analysis, technical reporting and literature review.